

P&ID SVG Inventory & Initial Analysis Report

Project: MINERVA — Flow Schemes and Instrumentation (P&ID reproduction to ISO / ISA 5.1)

Generated: 2026-06-03

Source archives:

- 02 - Flow schemes and Instrumentation.zip (≈ 0.97 MB)
 - 02 - Flow schemes and Instrumentation_Master.zip (≈ 72.6 MB)
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1. Extraction Summary

Both archives were unzipped into organized directories:

```
/home/ubuntu/pid_project/
├── extracted/
│   ├── standard/      # contents of "02 - Flow schemes and Instrumentation.zip"
│   └── master/        # contents of "...Instrumentation_Master.zip"
├── svg_source/        # de-duplicated working copies of the 2 unique SVGs
├── previews/          # rendered PNG previews of each SVG
├── analyze_svg.py     # analysis script
└── svg_inventory.md   # this report
```

Archive contents

File	standard.zip	master.zip	Type
PFD-PID MINERVA QCELL-LB.svg	✓	✓	SVG (P&ID)
PFD-PID MINERVA RF- CELL seen by ACR.svg	✓	✓	SVG (P&ID)
MANIFEST_02 - Flow schemes and Instru- mentation.html	✓	✓	Manifest
PID Nomenclature MINERVA CryoCell (QCELL-LB).xlsx	✓	✓	Tag nomenclature
QCELL - Auxilliary lines (NA.CP).xlsx	—	✓	Auxiliary line list
2024-03-07 - Brain- storming QM instru- mentation and con- trols.pptx	—	✓	Reference
PFD-PID of RFCELL - MASTER.pptx	—	✓	Source/master deck
PID MINERVA CryoCell (QCELL-LB).pptx	—	✓	Source deck
QSYS (and RFCELL) instrumentation loca- tion for LB and LBI.pptx	—	✓	Reference (65 MB)

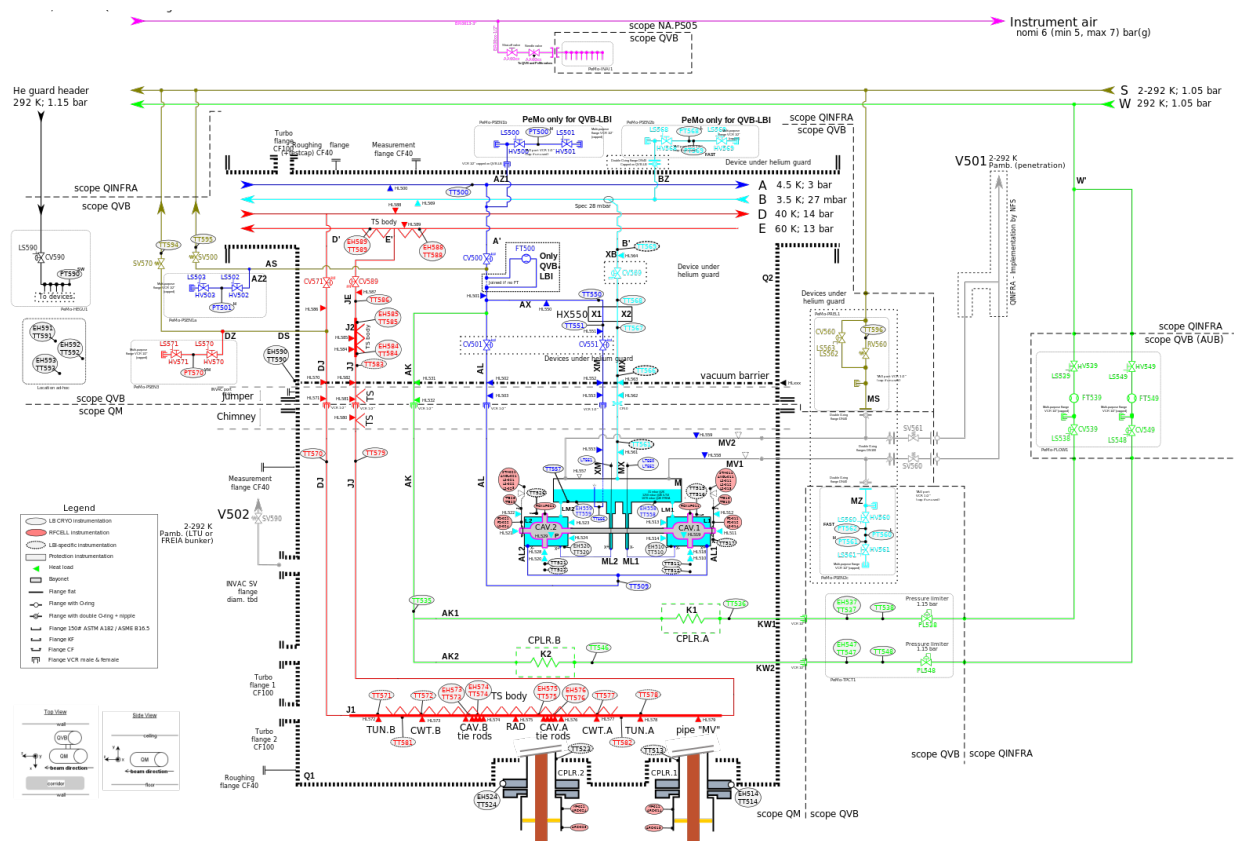
Note: The two SVG files are byte-for-byte identical across both archives (verified by MD5). The `_Master.zip` additionally contains the editable PowerPoint sources and supporting Excel nomenclature/auxiliary-line lists, which will be valuable references for the standards-compliant reproduction.

Unique SVG files found: 2

#	File name	MD5	Size
1	PFD-PID MINERVA QCELL-LB.svg	788e30df...99e	1.70 MB
2	PFD-PID MINERVA RF- CELL seen by ACR.svg	d7fc4243...99b	0.96 MB

2. SVG #1 — PFD-PID MINERVA QCELL-LB.svg

The primary, most detailed P&ID — the full QCELL / CryoCell cryogenic flow scheme with complete instrumentation.



Document properties

Property	Value
Internal docname	PFD_MINERVA_QCELL-LB.svg
Authoring tool	Inkscape 1.4 (2024-10-11)
SVG version	1.1
viewBox	0 0 1527.2727 1080
Physical size	420 mm × 297 mm (ISO A3, landscape)
File size	1.70 MB

Structure & element counts

Element	Count
Total elements	3,463
path	1,029
tspan	663
g (groups)	586
text	459
marker (arrow/line-end defs)	336
circle	153
ellipse	118 (instrument bubbles)
rect	75
use	24
image	3 (raster insets)
pattern / clipPath	1 / 5

Inkscape layers — 63 layers (rich semantic segmentation already present)

The drawing is heavily layered. Key layer groups relevant to the segmentation goal:

- **Process / piping:** Piping and series instrumentation (×3), Piping , Bottom line and common , 6 - Main process line return + buffer tank (V) , 5 - Main process line inlet + buffer tank

(L) ,

4 - Coupler B intercept , 3 - Coupler A intercept + common

- **Items / components:** Vacuum vessel (×3), 2 K tank , CPLR 1 , CPLR 2 , CPLR circuit , CPLR cuff and port metal , CPLR pressure limiter , Bottom connection , Spare heaters , Flowmeter for AL , Headers A B C D
- **Instrumentation:** Instrumentation , Over-instrumentalization , Instrumentation Pcav Observer ,
Instrumentation naming zones , EH TT , EHTT
- **Sub-systems / zones:** RFCELL 1 , RFCELL 2 , INVAC (×3), QVB , QINFRA , QM - DUT at FREIA , QM - PeMo , He guard (×2), PeMo-HEGU1 , Instrument air , Utilities NA.PS , Users
- **Annotation:** Namings , Naming and conditions , Comments (×2), General comments , Heat loads (×2), Terminal points , Legend , Versions , Scope division , pn levels , electrical , TC-LB

Text / labels / tags

- **566 text strings** present (labels, tags, notes, legend).
- **203 unique ISA-style instrument tags.** Tag-prefix histogram:

Prefix	Meaning (ISA 5.1)	Count
TT	Temperature Transmitter	67
HL	Heat Load annotation	53
EH	Electric Heater	20
LS	Level Switch	17
HV	Hand Valve	12
CV	Control Valve	10
PT	Pressure Transmitter	9
SV	Safety / Solenoid Valve	6
CF	(CF-flange callouts)	6
LT	Level Transmitter	3
FT	Flow Transmitter	3
PL	Pressure Limiter	2
AA	Analysis/Alarm	2
RV	Relief Valve	1
HX	Heat Exchanger	1

- Sample component / equipment labels: CAV.1 , CAV.2 , CPLR.1 , CPLR.2 , CPLR.A , CPLR.B , HX550 , V501 , V502 , TUN.A/B , RAD , CWT.A/B , K1 , K2 , MV1 , MV2 .

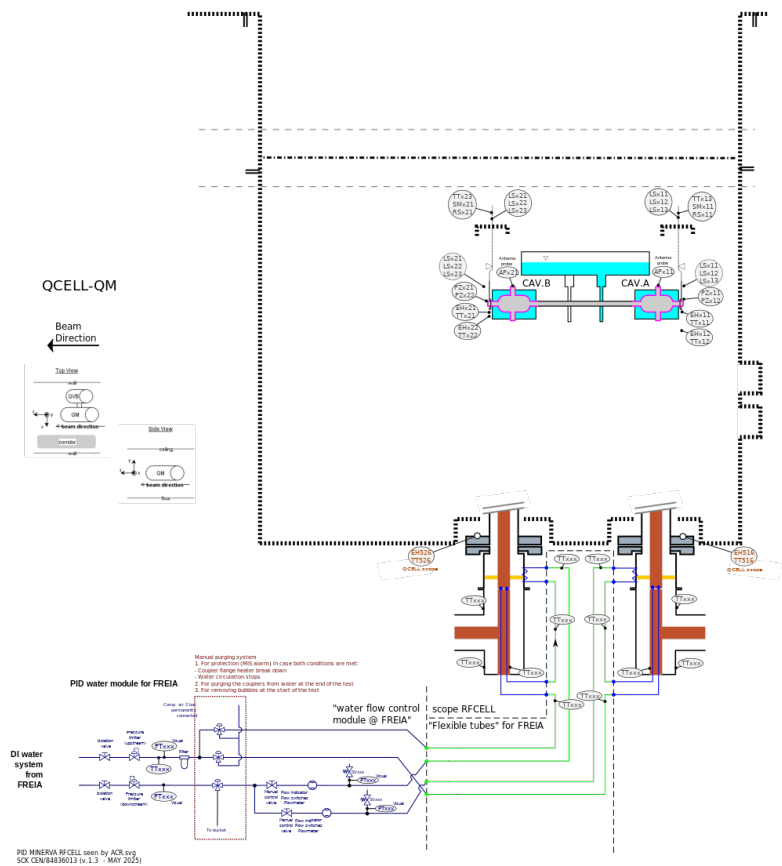
Color palette — 67 unique hex colors (process-line color coding present)

Color	Hex	Uses	Likely meaning
 Black	#000000	1144	Outlines, generic structure, text
 Blue	#0000ff	372	Process line A (4.5 K; 3 bar)
 Red	#ff0000	333	Process line D (40 K; 14 bar)
 Green	#00ff00	246	Coupler water / utility lines
 Cyan	#00ffff	194	Process line B (3.5 K; 27 mbar)
 Olive	#808000	160	He-guard header (292 K; 1.15 bar)
 Magenta	#ff00ff	151	Instrument air
 White	#ffffff	145	Fills
Dark grey	#1a1a1a	131	Text/strokes
Light grey	#f2f2f2 / #999999 / #808080	76 / 73 / 54	Devices-under-vacuum, shading
 Dark green	#008000	43	QINFRA scope lines
Navy	#000080	8	—
Salmon	#ffa000 / #d35f5f	13 / 8	RFCELL instrumentation bubbles
Amber	#ffcc00 / #f37f35	7 / 5	Highlights / heat

The legend in the drawing explicitly defines line classes **A 4.5 K/3 bar**, **B 3.5 K/27 mbar**, **D 40 K/14 bar**, **E 60 K/13 bar**, plus instrument-class fills (LB CRYO, RFCELL, LBI-specific, Protection) and a “**vacuum barrier**” dashed boundary — directly matching the requested segmentation axes.

3. SVG #2 — PFD-PID MINERVA RFCELL seen by ACR.svg

A focused P&ID of the RFCELL water-flow / coupler module (“as seen by ACR”), including the DI-water module for FREIA and the cavity/coupler detail.



Document properties

Property	Value
Internal docname	PID MINERVA RFCELL seen by ACR.svg
Reference	SCK CEN / 84836013 (v.1.3 — May 2025)
Authoring tool	Inkscape 1.4
SVG version	1.1
viewBox	0 0 1527.2727 1080
Physical size	420 mm × 297 mm (ISO A3, landscape)
File size	0.96 MB

Structure & element counts

Element	Count
Total elements	1,139
path	386
marker	221
tspan	166
g (groups)	126
text	78
circle	57
ellipse	39 (instrument bubbles)
rect	31
use	10
image	3
pattern / clipPath	2 / 9

Inkscape layers — 25 layers

- **Components:** CPLR A , CPLR B , Tee A , Tee B , coldmasses , CAV , CPLR , CPLR water loops , CPLR flange heater , Tubes for FREIA , Green pipes
- **Piping/process:** RFCELL , RF tee anti-freeze , Green pipes
- **Instrumentation:** Instrumentation , EH TT (×2), CPLR AD, ED, PT , RF Tee monitoring (FREIA)
- **Zones/annotation:** background , TUN , Legend , Comments , Comment , Scopes , Version

Text / labels / tags

- **128 text strings; 45 unique ISA-style tags.** Prefix histogram:

Prefix	Meaning (ISA 5.1)	Count
TT	Temperature Transmitter	23
LS	Level Switch	12
EH	Electric Heater	8
PT	Pressure Transmitter	6
PZ	Pressure (special)	4
AD	Antenna/Analysis device	4
SM	—	2
RS	—	2
AP	Antenna Probe	2
SV	Safety Valve	2
ED	—	2

- Note the **x placeholder** in tag names (e.g. TTx21 , EHx11 , LSx12 , PZx21) — this drawing uses a generic/templated numbering scheme for the two symmetric cavity strings (A/B).
- Equipment labels: CAV.A , CAV.B , QCELL-QM , plus notes on the manual purging system.

Color palette — 26 unique hex colors

Color	Hex	Uses	Likely meaning
 Black	#000000	453	Outlines, text
Navy	#000080	146	Water-loop / process line
 Blue	#0000ff	92	Process line
Dark grey	#4d4d4d	65	Structure
 White	#ffffff	64	Fills
 Red	#ff0000	59	Heater / hot line
 Green	#00ff00	48	Coupler water lines (to FREIA)
Light grey	#f2f2f2	43	Shading
 Dark green	#008000	34	Green pipes / scope
 Cyan	#00ffff	24	Cold line
Brown	#aa4400 / #bf512e	10 / 7	Cavity/coupler bodies
Amber	#ffcc00 / #ff9a00	2 / 1	Highlights

4. Cross-Cutting Observations (relevant to the reproduction goal)

The requested segmentation axes map directly onto structures **already present** in the source SVGs, which makes a clean, standards-compliant rebuild feasible:

Requested segmentation	Source signal already in the SVGs
Main lines	Color-coded line classes A/B/D/E + dedicated Piping... , Main process line... layers
Individual items	Equipment layers: Vacuum vessel , 2 K tank , CPLR 1/2 , CAV , HX550 , V501/V502
Components	CPLR circuit , CPLR cuff and port metal , CPLR water loops , Bottom connection
Sensors	ellipse / circle instrument bubbles (118 / 39) + Instrumentation layers
Instrumentation tags	203 + 45 unique ISA-style tags (TT , PT , LT , LS , FT , EH , CV , HV , SV , RV)
Vacuum barrier	Explicit dashed “ vacuum barrier ” boundary + INVAC layers
Temperature	TT / LT tags, EH heaters, Heat loads / HL annotations, cryo temperatures in legend
Pressure	PT / PZ / PL / RV tags, pressure values on each line class in legend
Color segmentation	67 / 26 hex colors; line classes and instrument classes already color-keyed

Standards notes for the rebuild

- Both sheets are **ISO A3 landscape** with the same viewBox (0 0 1527.2727 1080), so a shared coordinate grid / template can be reused.
- Tags broadly follow **ISA 5.1** letter conventions (first letter = measured variable, e.g. T, P, L, F; trailing function letters T = transmitter, V = valve, S = switch). A normalization pass against the PID Nomenclature MINERVA CryoCell (QCELL-LB).xlsx will be needed to make tags fully ISA-consistent.
- Instrument **bubbles** are drawn as ellipse / circle ; line-end **markers** (336 / 221) are heavily used for arrowheads, heat-load triangles, and flange symbols — these will need a clean symbol library.
- Color usage is currently RGB-primary (#ff0000 , #00ff00 , #0000ff , ...). For an ISO/ISA reproduction a defined, documented color palette should replace the ad-hoc primaries.

5. Inventory at a glance

#	SVG	Sheet size	view-Box	Elements	Layers	Text	Unique tags	Colors
1	PFD-PID MIN- ERVA QCELL- LB	A3 (420×2 97 mm)	0 0 1527.27 1080	3,463	63	566	203	67
2	PFD-PID MIN- ERVA RFCELL seen by ACR	A3 (420×2 97 mm)	0 0 1527.27 1080	1,139	25	128	45	26

Total: 2 unique P&ID SVG files, both Inkscape-authored ISO A3 landscape drawings sharing a common

coordinate system, with existing semantic layering and color coding that align well with the planned ISO / ISA 5.1 segmentation and reproduction work.